

# •Fiber Optic Network Design

College of Opt. Sci. & Engr., ZJU

2026

# •Outline

## •Optical Amplification

### •EDFA

- Structure
- Parameters
- Selection
- Gain flattening

### •Exercise

# • Why optical amplifier

An **optical amplifier** is a device which amplifies the optical signal directly without ever changing it to electricity. The light itself is amplified.

Reasons to use the optical amplifiers:

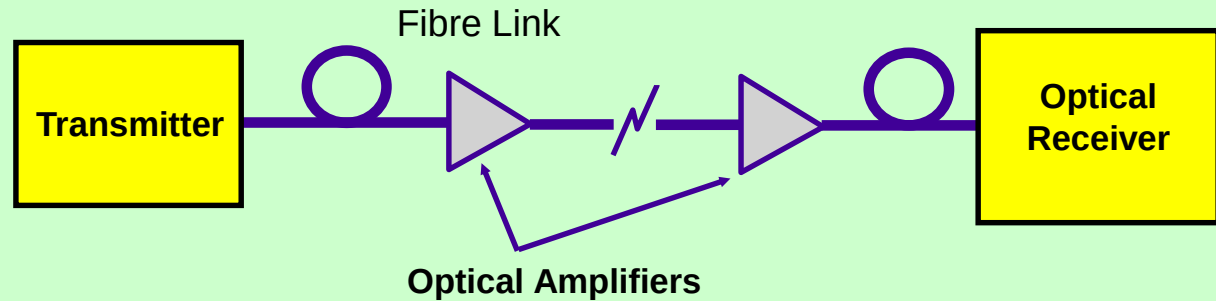
- Reliability
- Flexibility
- Wavelength Division Multiplexing (WDM)
- Low Cost

# Optical Amplification

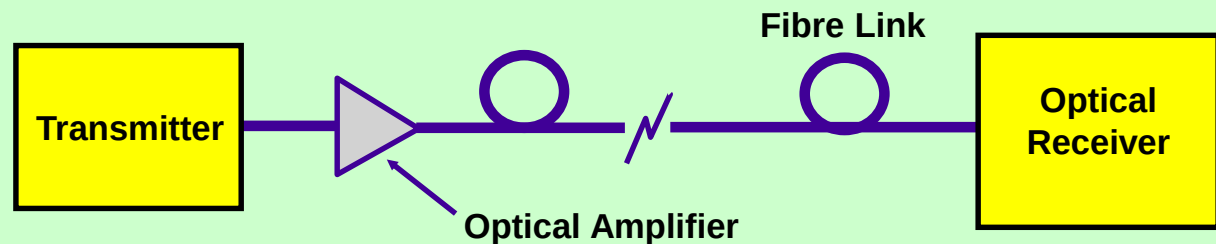
- Variety of optical amplifier types exist, including:
  - Semiconductor optical amplifiers
  - Optical fiber amplifiers (Erbium Doped Fiber Amplifiers)
  - Distributed fiber amplifiers (Raman Amplifiers)
- Optical fiber amplifiers are now the most common type
- One of the most successful optical processing functions
- Also used as a building block in DWDM systems

# Optical Amplifier Applications

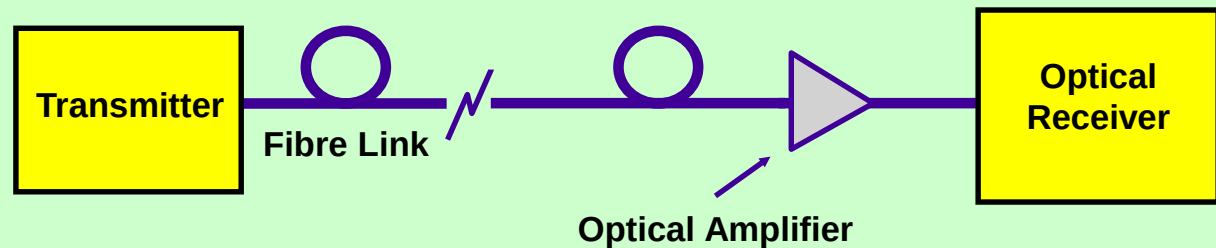
**In-line  
Amplifier**



**Power  
Amplifier**



**Preamplifier**



# Amplifier Applications

## *Preamplifiers*

- a very low noise characteristic is essential
- only a moderate gain figure
- Typically a preamplifier will not have feedback control as it can be run well below saturation.

## *Power amplifiers*

- Most DFB lasers output : 2 mW,
- A fiber: aggregate power levels of up to 100 to 200 mW before nonlinear effects start to occur.
- Typical EDFA power amplifiers have an output of around 100 mW

## *Line amplifiers*

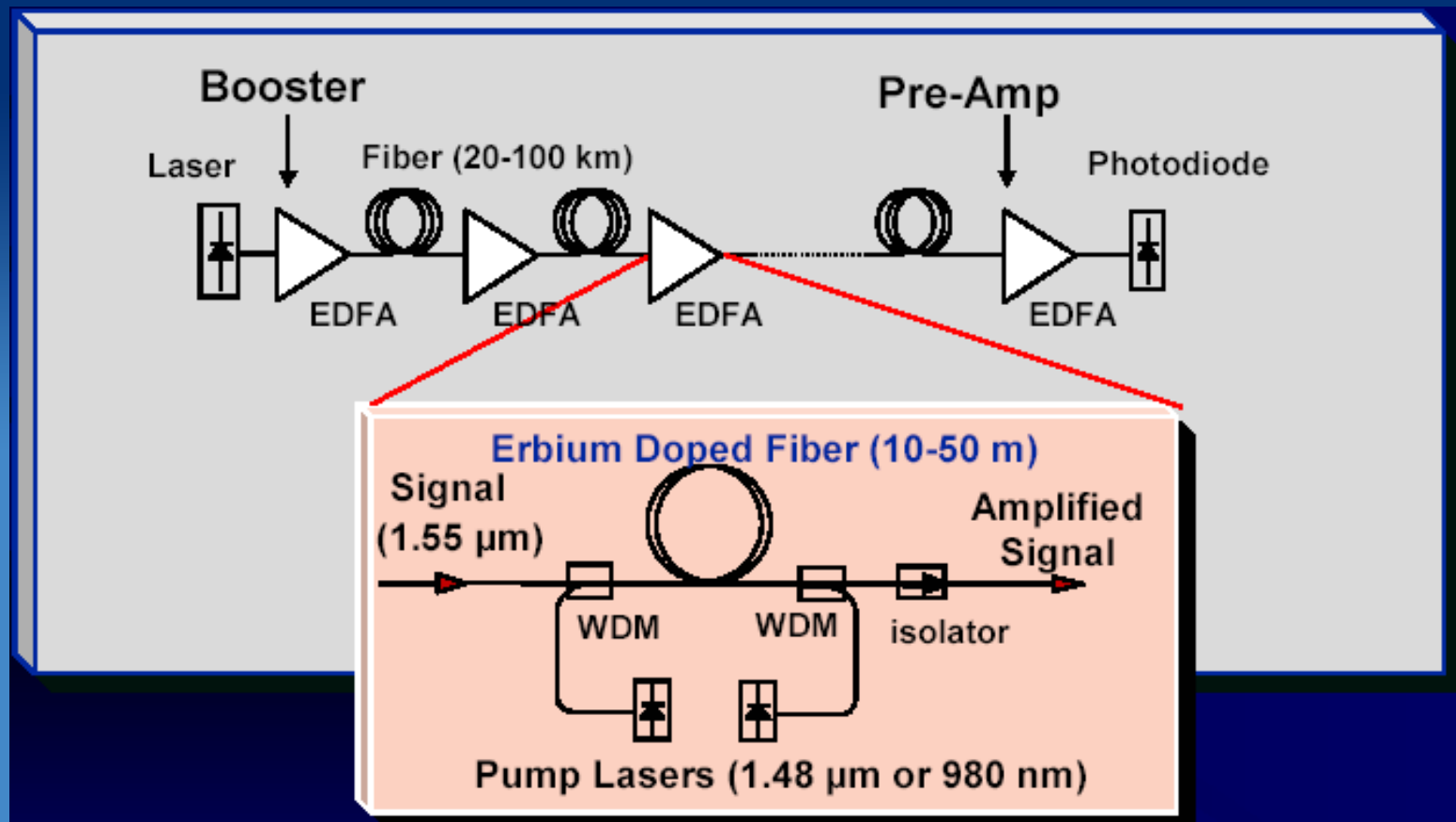
- Replaces a repeater within a long communication line
- high gain at the input and high power output
- a very low noise figure.
- This is really a preamplifier cascaded with a power amplifier. Sophisticated line amplifiers today tend to be made just this way - as a multi-section amplifier separated by an isolator.

# Selecting Amplifiers

| Type            | Gain        | Maximum Output power | Noise figure               |
|-----------------|-------------|----------------------|----------------------------|
| Power Amplifier | High gain   | High output power    | Not very important         |
| In-line         | Medium gain | Medium output power  | Good noise figure          |
| Preamplifier    | Low gain    | Low output power     | Low value < 5 dB essential |

# Optically Amplified Systems

**EDFA = Erbium Doped Fiber Amplifier**



Bits continue in photonic format



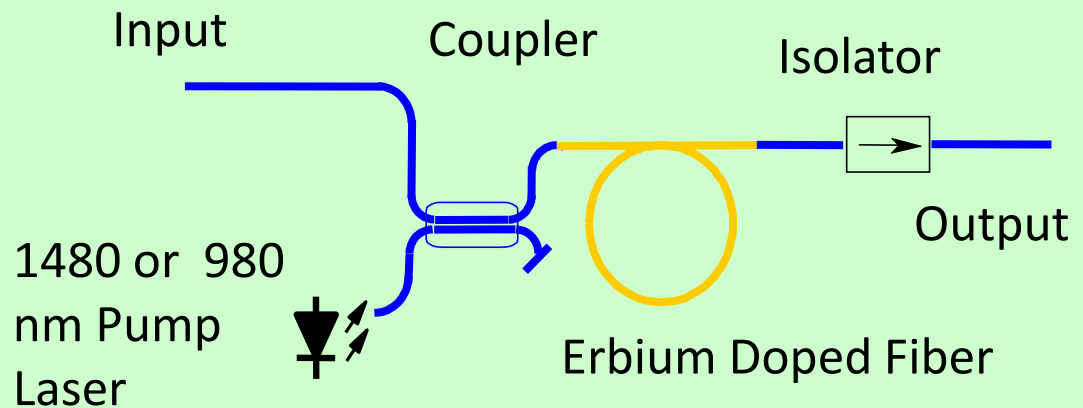
# Overview

- **Erbium doped fiber amplifiers**
- **Issues: Noise and gain flattening**

# Basic EDF Amplifier Design

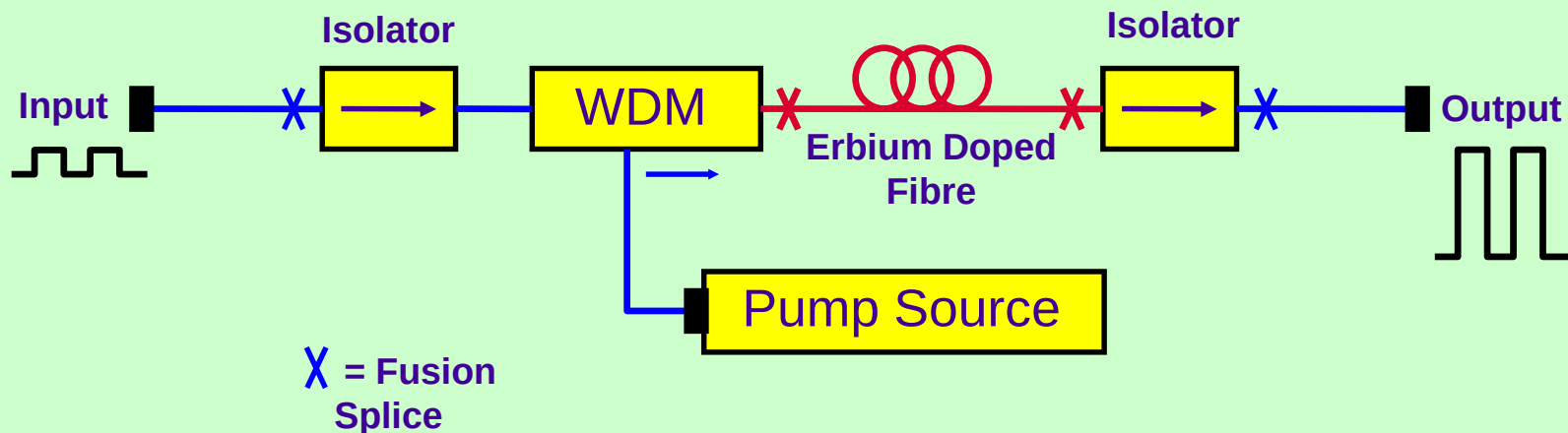
- Erbium-doped fiber amplifier (EDFA) most common
  - Invented at Stanford in early 1980s by H. J. Shaw and Michel Digonnet
  - Commercially available since the early 1990's
  - Works best in the range from 1530 to 1565 nm
  - Gain up to 30 dB (1000 photons out per photon in!)

- Optically transparent
  - “Unlimited” RF bandwidth
  - Wavelength transparent

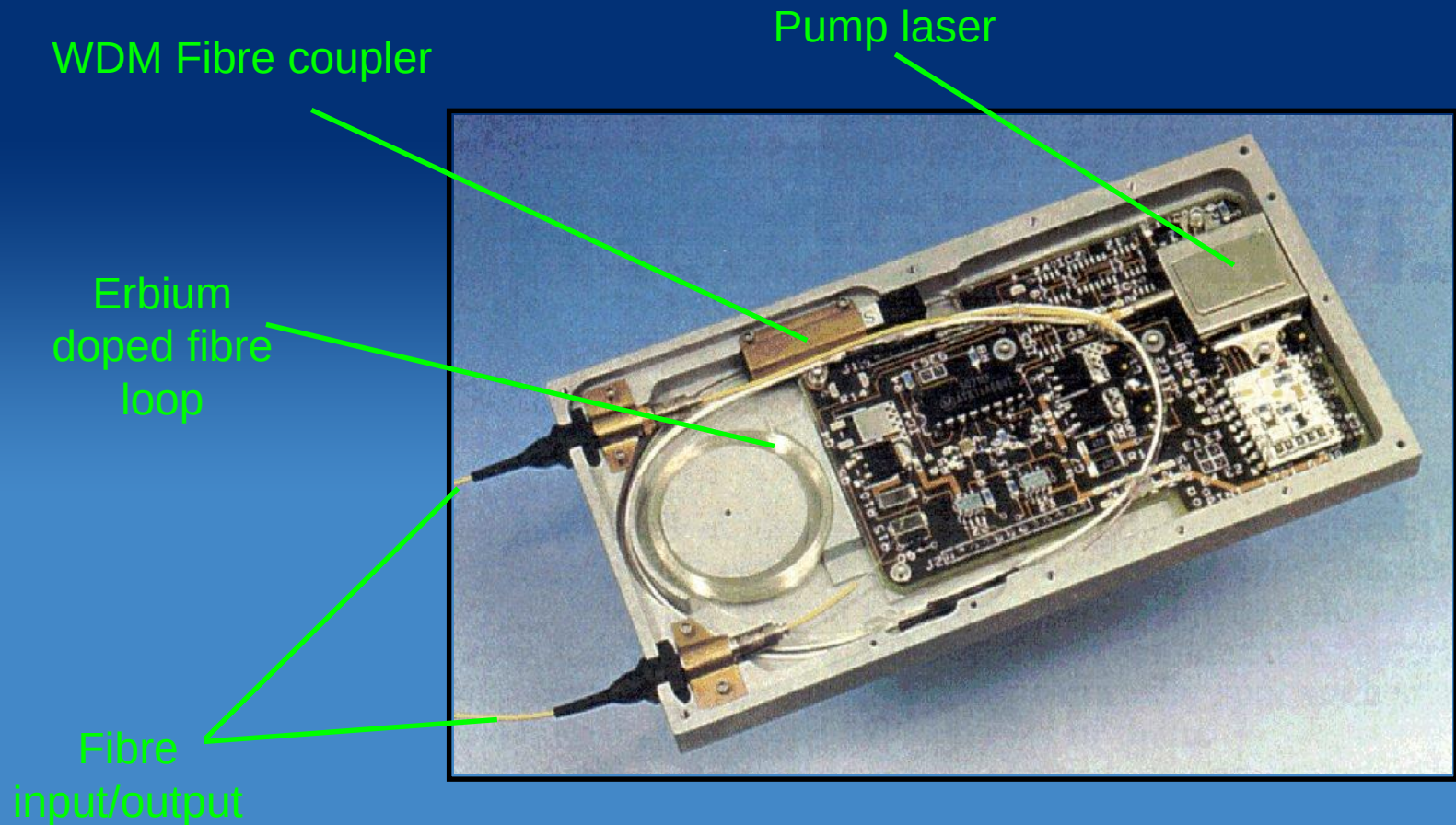


# Erbium Doped Fiber Amplifier

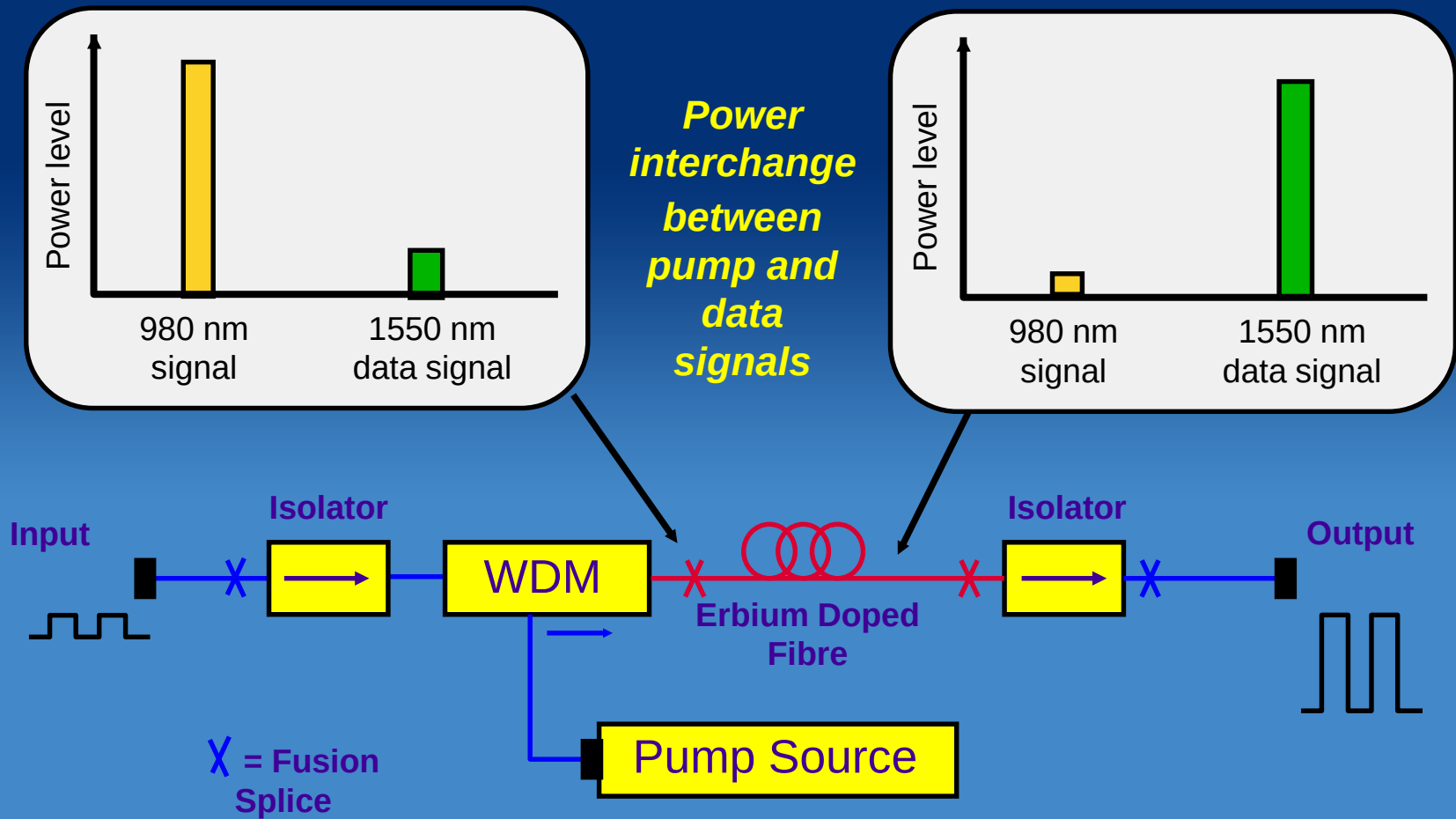
- A pump optical signal is added to an input signal by a WDM coupler
- Within a length of doped fiber part of the pump energy is transferred to the input signal by stimulated emission
- For operation  $\sim 1550$  nm the fiber dopant is Erbium
- Pump wavelength is either 980 nm or 1480 nm, pump power  $\sim 50$  mW
- Gains of 30-40 dB possible



# Interior of an Erbium Doped Fibre Amplifier (EDFA)

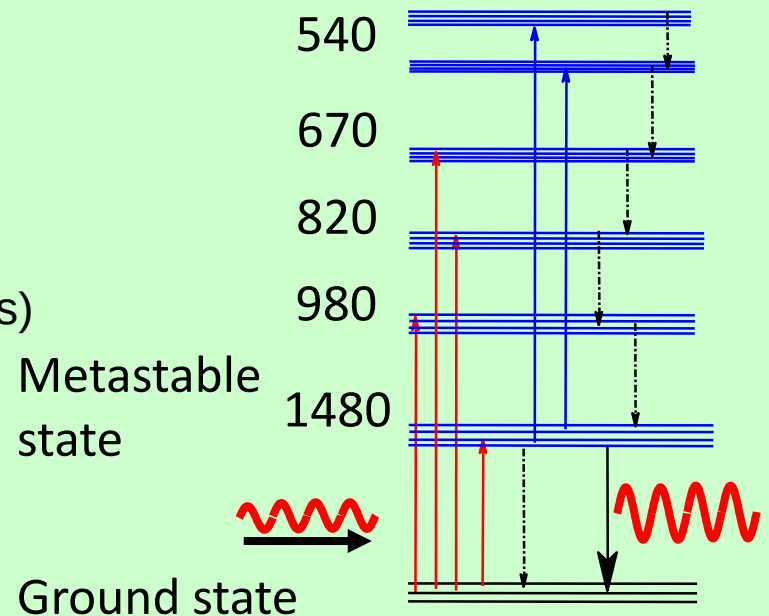


# Operation of an EDFA

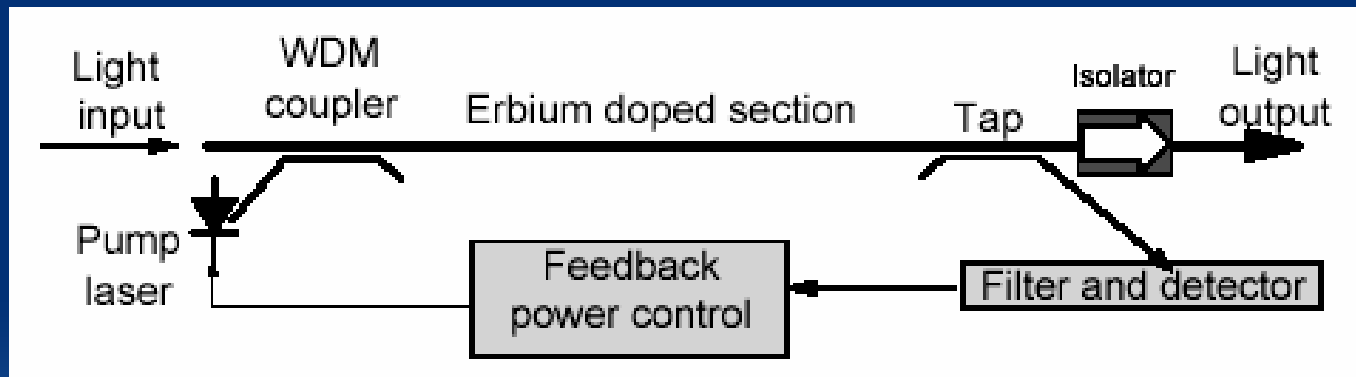


# Erbium Properties

- Erbium: rare element with phosphorescent properties
  - Photons at 1480 or 980 nm activate electrons into a metastable state (why these two pumping wavelengths? No excitation level absorption!)
  - Electrons falling back emit light in the 1550 nm range
- Spontaneous emission
  - Occurs randomly (time constant ~1 ms)
- Stimulated emission
  - By electromagnetic wave
  - Emitted wavelength & phase are identical to incident one



# Erbium Doped Fiber Amplifiers



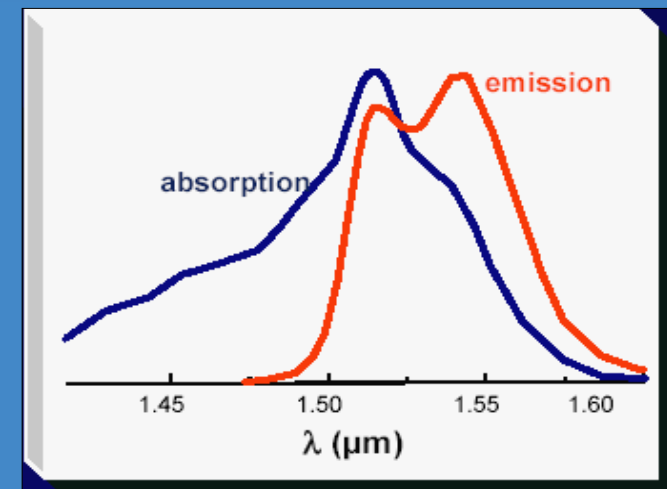
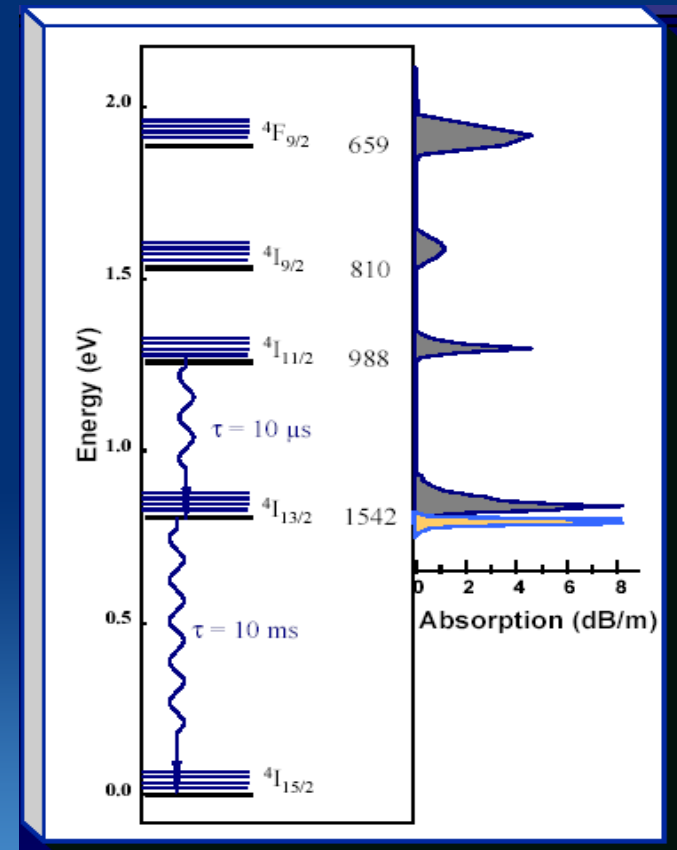
- Fiber length: typically ten metres or so
- Doping: a small controlled amount of the rare earth element erbium added to the glass in the form of an ion ( $\text{Er}^{3+}$ ).
- LD power: between 10 and 200 mW
- Light wavelength at either 980 or 1,480 nm.

Question #1: Why not increase doping level and decrease length?

Question #2: What's the advantage of pumping 980 nm or 1480nm?

# Er<sup>3+</sup> Energy Levels

- Pump:
  - 980 nm (low noise) or 1480 nm (high efficiency)
  - Pump power >5 mW
  - The 980 nm band: high absorption cross-section, generally used where low-noise performance is required. The absorption band is relatively narrow and so wavelength stabilized laser sources are typically needed.
  - The 1480 nm band: higher, but broader absorption cross-section, generally used for higher power amplifiers.
  - A combination of 980 nm and 1480 nm pumping is generally utilized in amplifiers.
- Emission:
  - 1.52-1.57  $\mu\text{m}$
  - Long living upper state (10 ms)
  - Gain  $\cong$  30 dB





# Technical Characteristics of EDFA

**EDFAs have a number of attractive technical characteristics:**

- **Efficient pumping**
- **Minimal polarisation sensitivity**
- **Low insertion loss**
- **High output power (this is not gain but raw amount of possible output power)**
- **Low noise**
- **Very high sensitivity**
- **Low distortion and minimal interchannel crosstalk**

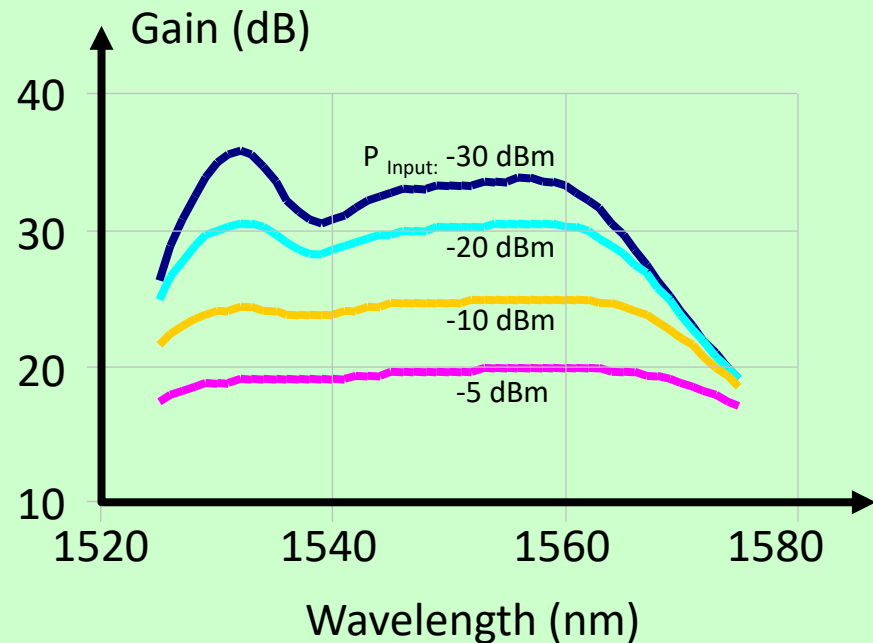
# Optical Gain (G)

- $G = S_{\text{Output}} / S_{\text{Input}}$

$S_{\text{Output}}$ : output signal (without noise from amplifier)

$S_{\text{Input}}$ : input signal

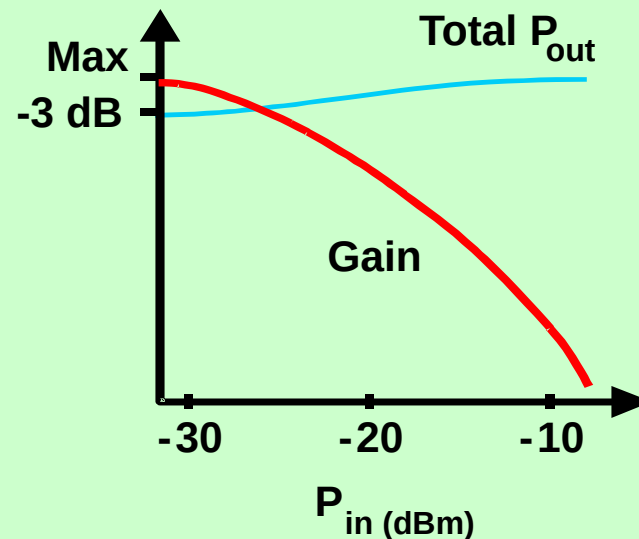
- Input signal dependent
  - Operating point (saturation) of EDFA strongly depends on **power** and **wavelength** of incoming signal



# Saturation in EDFAs

Total output power:

Amplified signal + Noise (Amplified Spontaneous Emission ASE)



EDFA is in saturation if almost all Erbium ions are consumed for amplification  
Total output power remains almost constant, regardless of input power changes

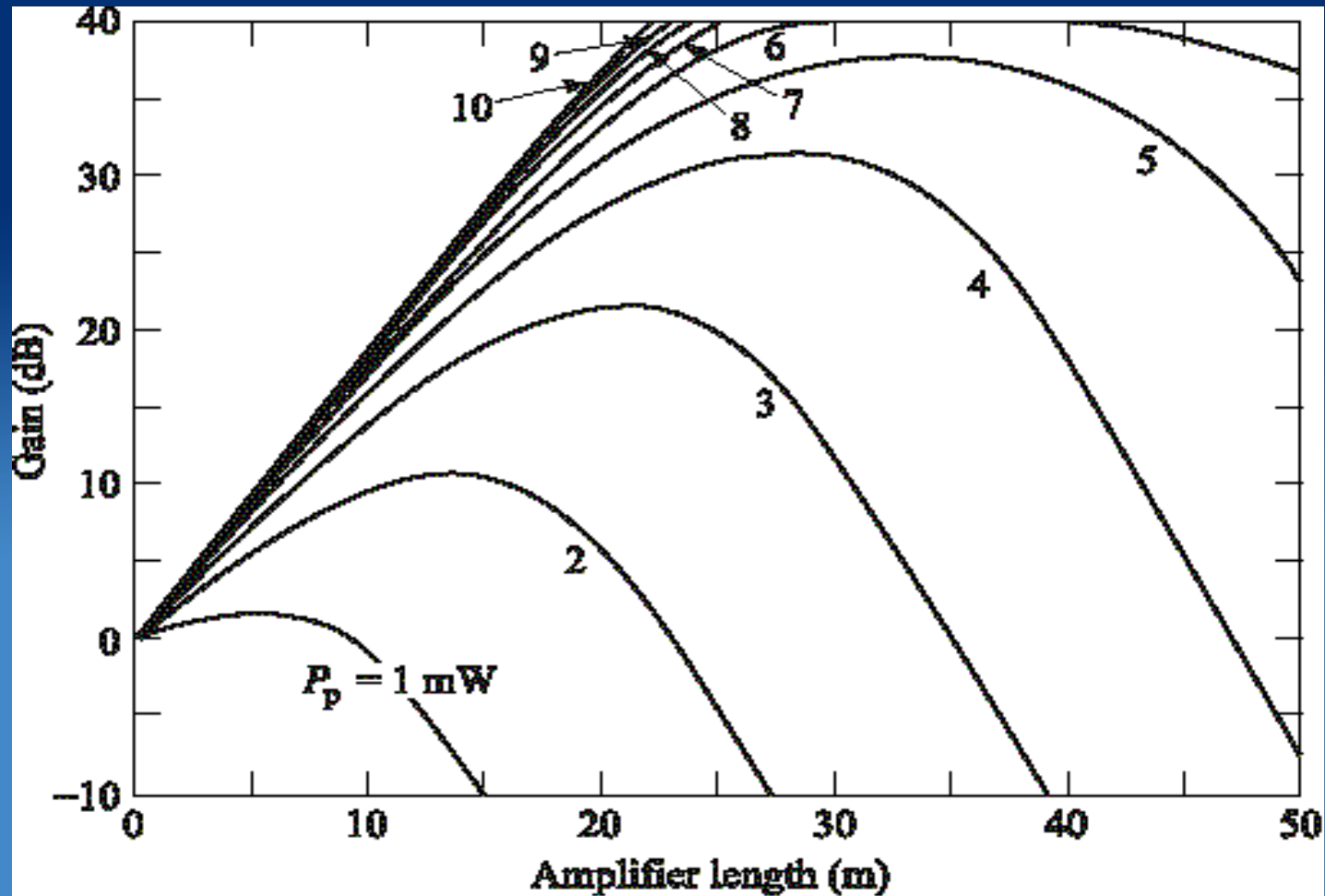
# Amplifier Length

- As the signal travels along the length of the amplifier it becomes stronger due to amplification.
- As the pump power travels through the amplifier its level decreases due to absorption.
- Thus, both the signal power level and the pump power level vary along the length of the amplifier. At any point we can have only a finite number of erbium ions and therefore we can only achieve a finite gain (and a finite maximum power) per unit length of the amplifier.

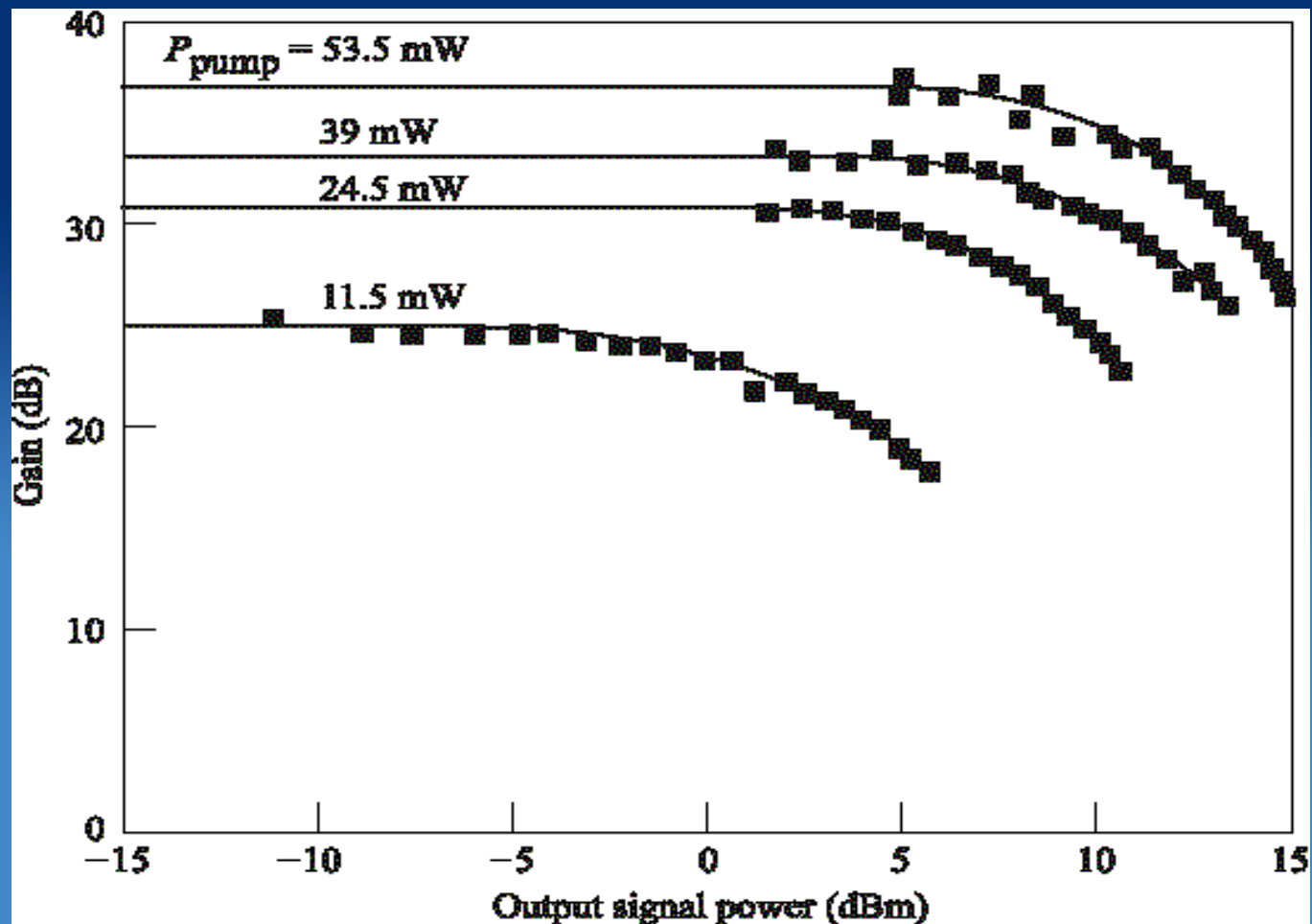
**Fiber length=F(signal power, pump power, Er concentration, gain required)**

- In an amplifier designed for multiwavelength operation there is another consideration - the flatness of the gain curve over the range of amplified wavelengths. With a careful design and optimisation of the amplifier's length we can produce a nearly flat amplifier gain curve.

# Gain versus 'optimum' fiber length



# Gain versus pump level

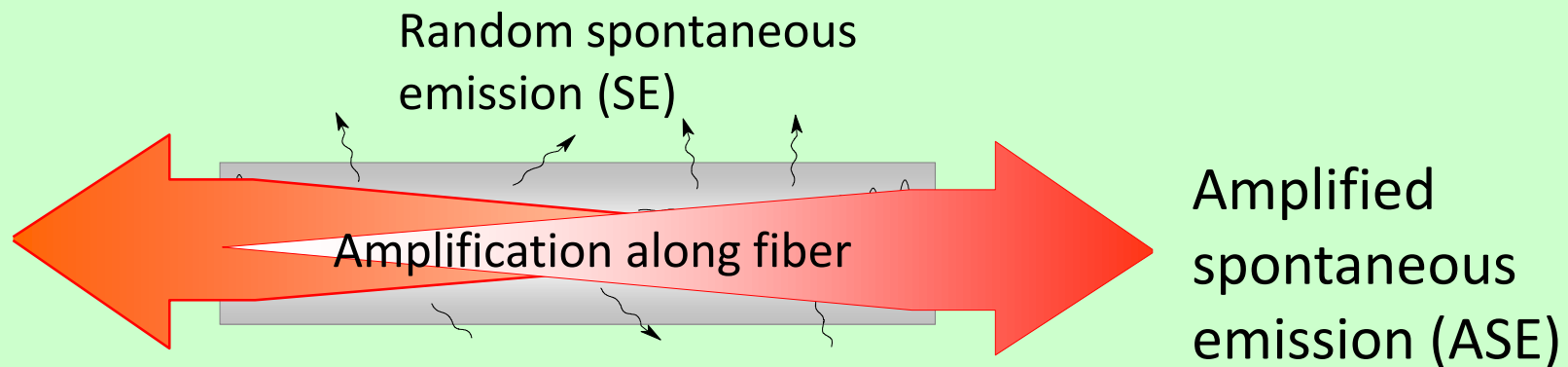


# Noise of (E)DFAs

- The principal source of noise in DFAs is Amplified Spontaneous Emission (ASE), which has a spectrum approximately the same as the gain spectrum of the amplifier.
- Noise figure in an ideal DFA is 3 dB.
- Practical amplifiers can have noise figure as large as 6–8 dB.

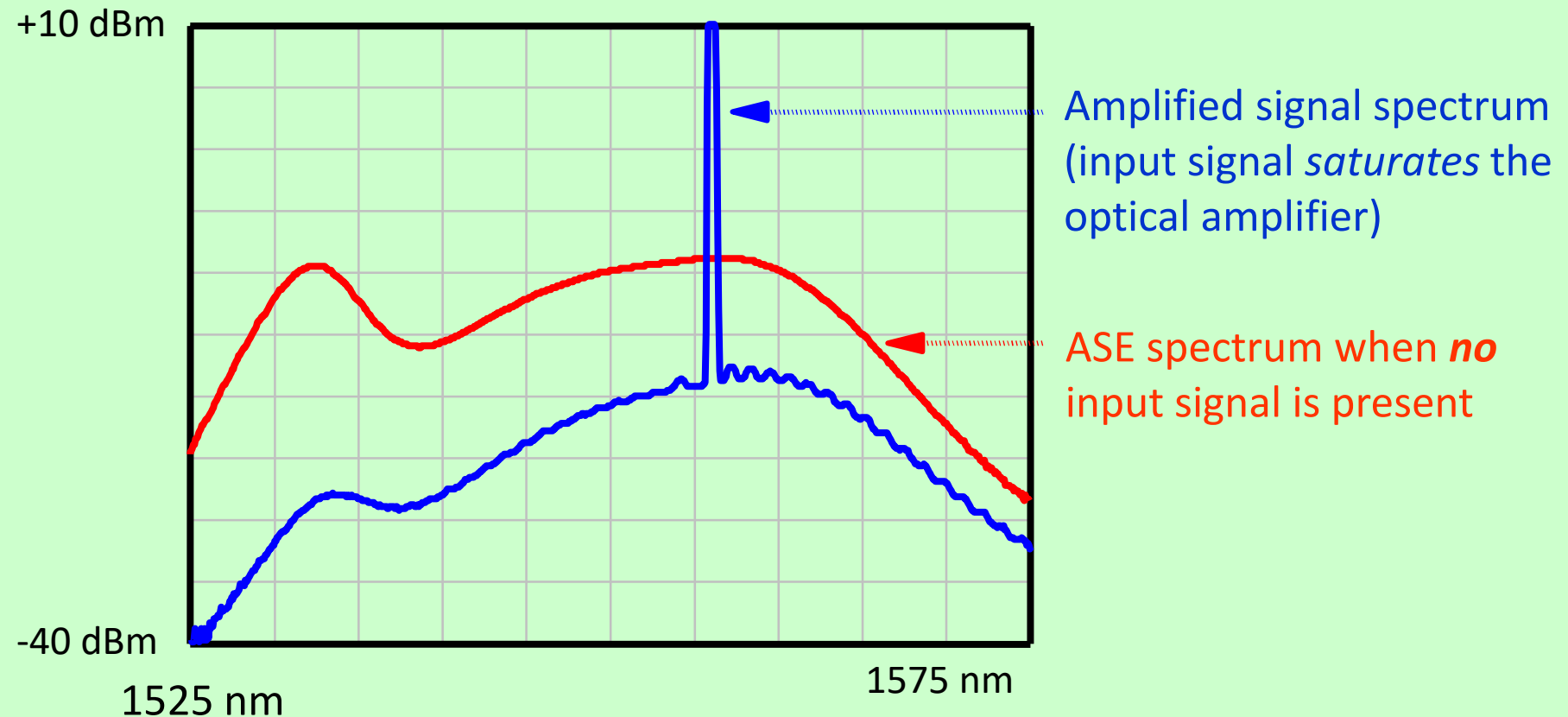
# Amplified Spontaneous Emission

- Erbium randomly emits photons between 1520 and 1570 nm
  - Spontaneous emission (SE) is not polarized or coherent
  - Like any photon, SE stimulates emission of other photons
  - With no input signal, eventually all optical energy is consumed into amplified spontaneous emission
  - Input signal(s) consume metastable electrons → much less ASE



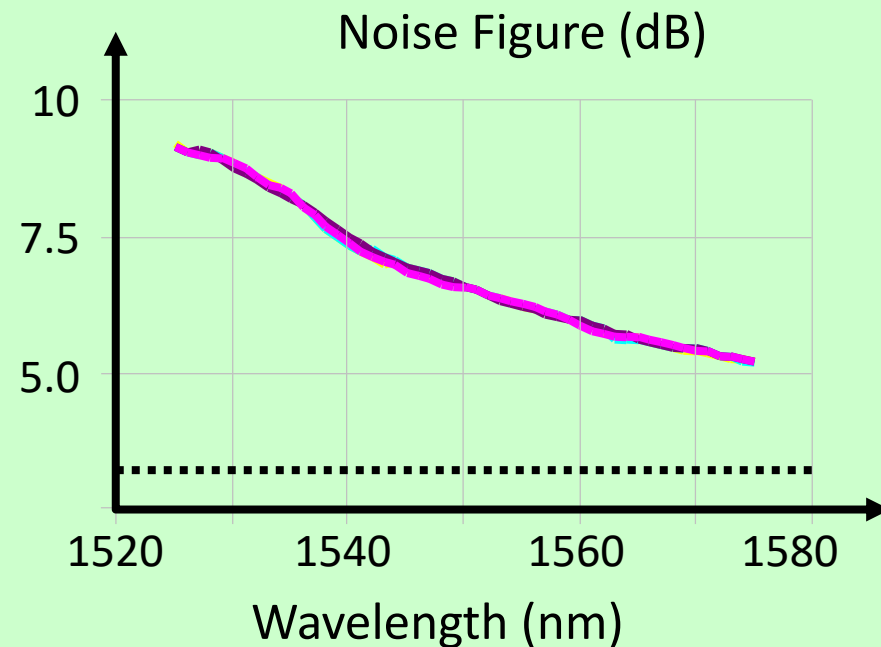


# Output Spectra

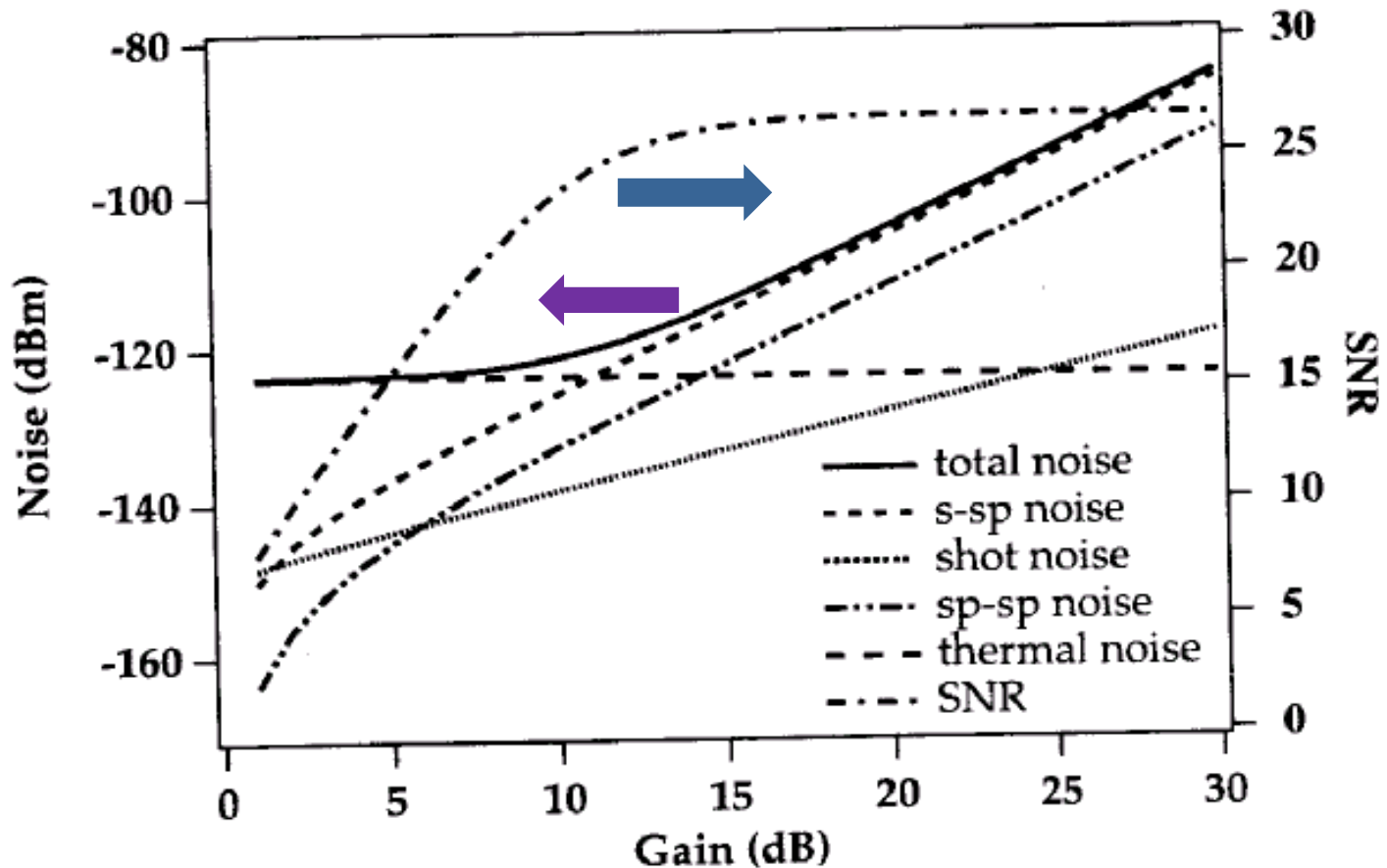


# Noise Figure (NF)

- $NF = P_{ASE} / (h \cdot \nu \cdot G \cdot B_{OSA})$ 
  - $P_{ASE}$ : ASE power measured by OSA
  - $h$ : Planck's constant
  - $\nu$ : Optical frequency
  - $G$ : Gain of EDFA
  - $B_{OSA}$ : Optical bandwidth [Hz] of OSA
- Input signal dependent
  - In a saturated EDFA, the NF depends mostly on the wavelength of the signal
  - Physical limit: 3.0 dB



# Noise Sources



Note: sp-sp=自发辐射-自发辐射差拍噪声, s-sp=signal-自发辐射差拍噪声

Ref: <https://max.book118.com/html/2015/0305/12982427.shtml>

# EDFA

- EDFA has revolutionized optical communications
  - All optical and fiber compatible
  - Wide bandwidth, 20~70 nm
  - High gain, 20~40 dB
  - High output power, >200mW
  - Bit rate, modulation format, power and wavelength insensitive
  - Low distortion and low noise ( $NF < 5\text{dB}$ )

# Example: Conventional EDFA

- Best used for single channel systems in the 1550 nm region,
- Systems are designed for use as boosters, in-line amplifiers or preamplifiers.
- Bandwidth is not wide enough for DWDM, special EDFA needed

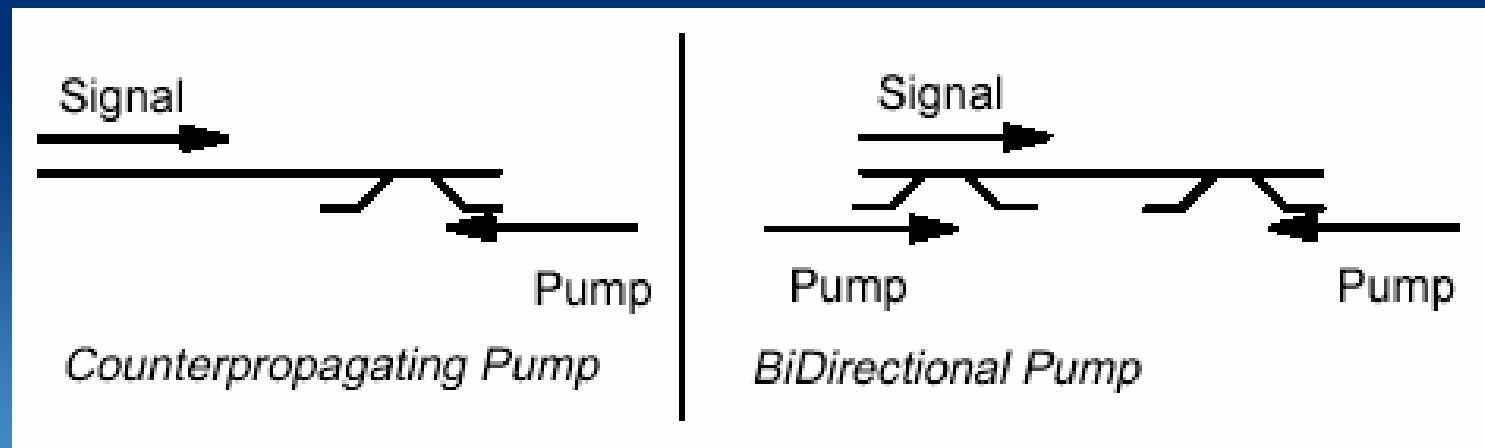
| Erbium Doped Fibre Amplifier        |     | EDFA-14   | EDFA-16   | EDFA-19   |
|-------------------------------------|-----|-----------|-----------|-----------|
| Wavelength Range                    | nm  | 1525-1565 | 1525-1565 | 1525-1565 |
| Saturation Power                    | dBm | 14        | 16        | 19        |
| Maximum Gain in Small Signal        | dB  | >35       | >38       | >40       |
| Optimum Gain Flatness (1545-1558nm) | dB  | 0.6       | 0.6       | 0.6       |
| Noise Figure                        | dB  | <6        | <6.5      | <6        |
| Min. Isolation at Input and Output  | dB  | 30        | 30        | 30        |
| Connectors                          |     | FC/APC    | FC/APC    | FC/APC    |

# Gain Flattened EDFA for DWDM

- Gain flatness is now within 1 dB from 1530-1560 nm
- ITU-T DWDM C band is 1530 to 1567 nm

| Flat Gain EDFA                      |     | EDFA-FG13 | EDFA-FG15 | EDFA-FG18 |
|-------------------------------------|-----|-----------|-----------|-----------|
| Wavelength Range                    | nm  | 1530-1560 | 1530-1560 | 1532-1560 |
| Saturation Power                    | dBm | 13        | 15        | 18        |
| Gain Value                          | dB  | >20       | >22       | >25       |
| Optimum Gain Flatness (1530-1560nm) | dB  | <1        | <1        | <1        |
| Noise Figure                        | dB  | <6.0      | <6.5      | <6.0      |
| Min. Isolation at Input and Output  | dB  | 30        | 30        | 30        |
| Connectors                          |     | FC/APC    | FC/APC    | FC/APC    |

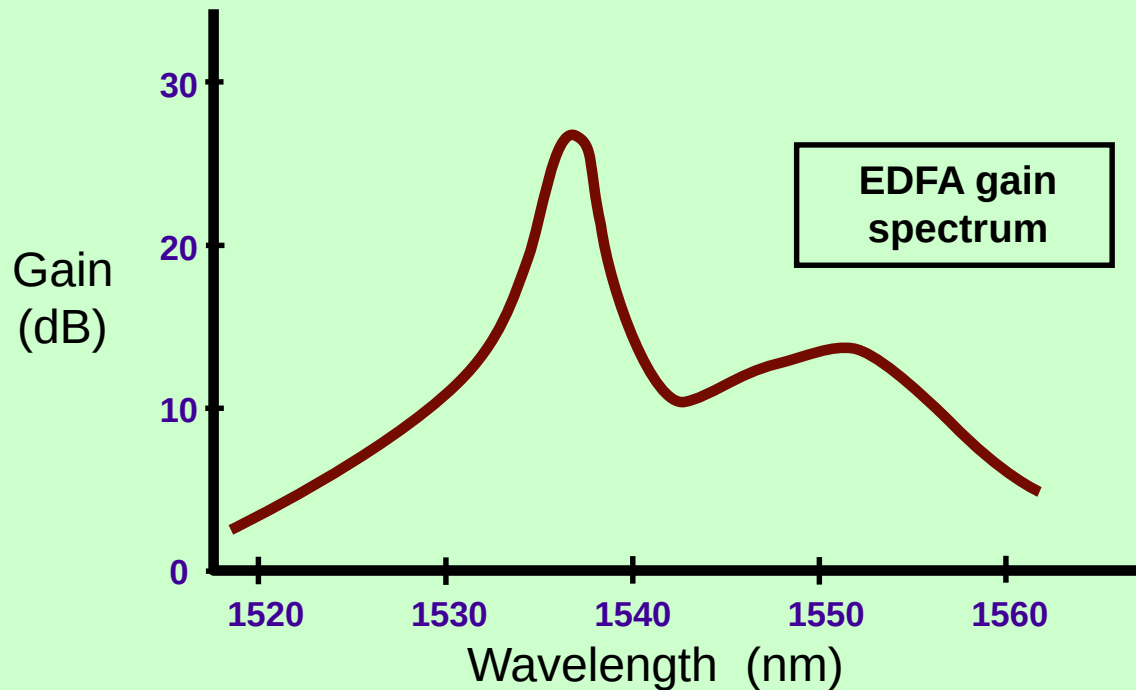
# Pumping Directions



***Additional pumping options***

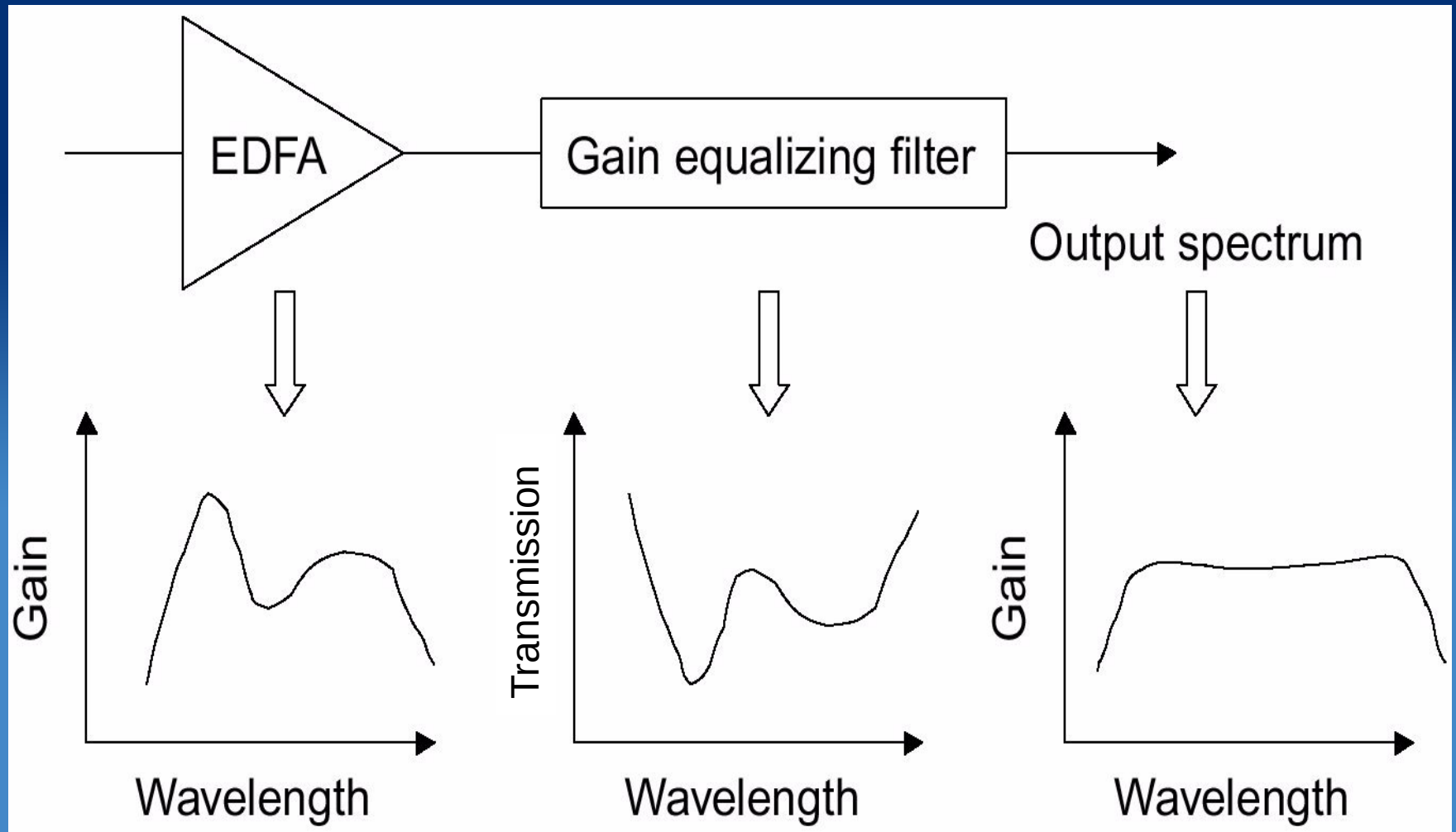
# EDFA Gain Spectrum

- Erbium can provide about 40-50 nm of bandwidth, from 1520 to 1570 nm
- Gain spectrum depends on the glass used, eg. silica or zblan glass
- Gain spectrum is not flat, significant gain variations





# Gain Flattening Concept



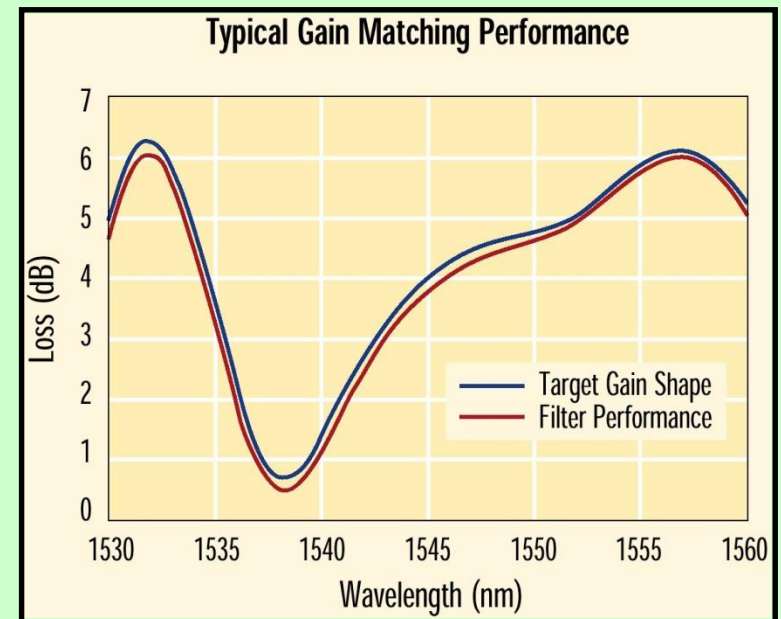
# Gain Flattening Filters (Equalizers)

- Used to reduce variation in amplifier gain with wavelength, used in DWDM systems
- The gain equalisation is realised by inserting tapered long period gratings within the erbium doped fibre.
- Designed to have approximately the opposite spectral response to that of an EDFA

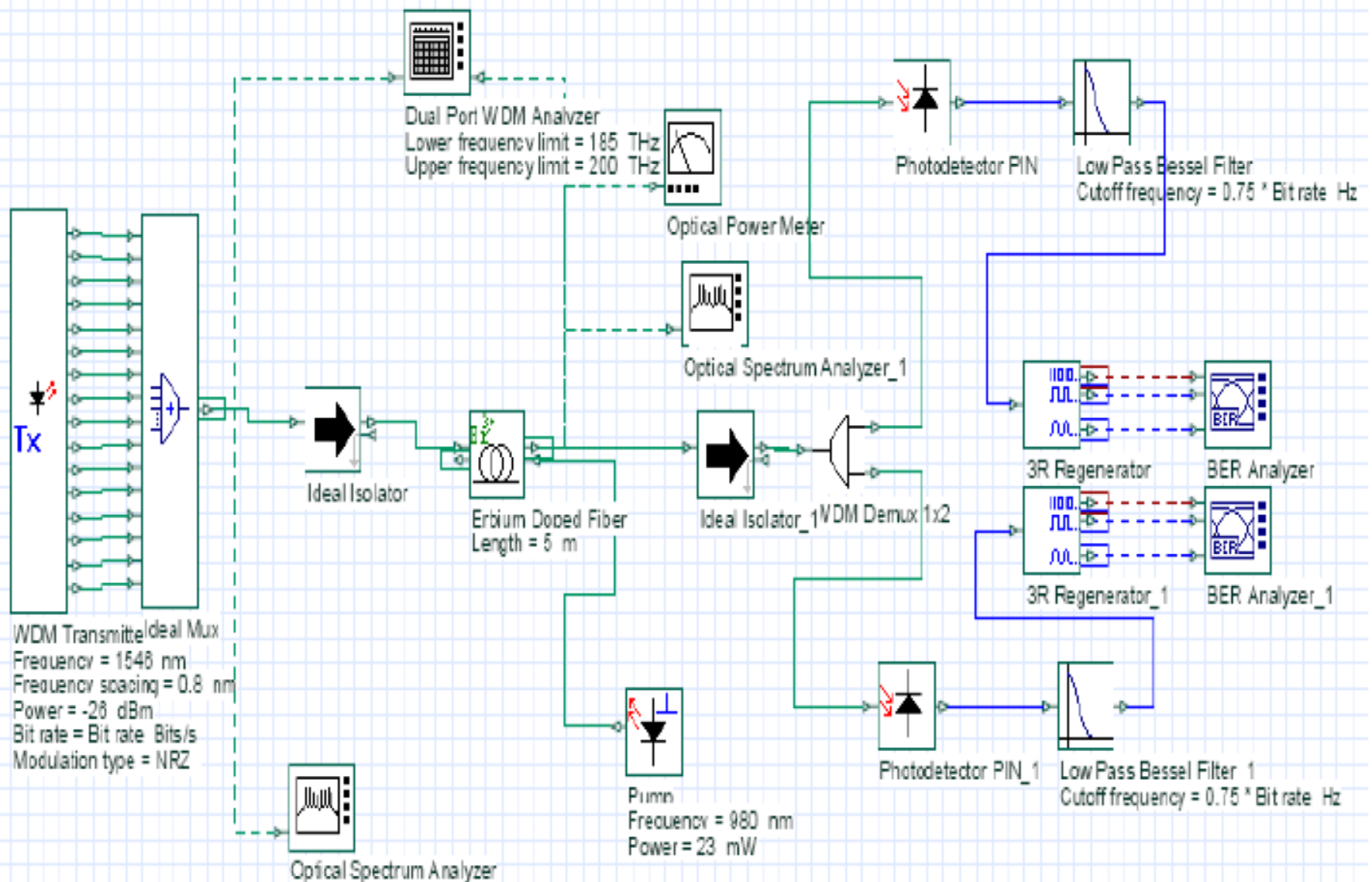
Inline Dicon gain  
flattening filter spectral  
response



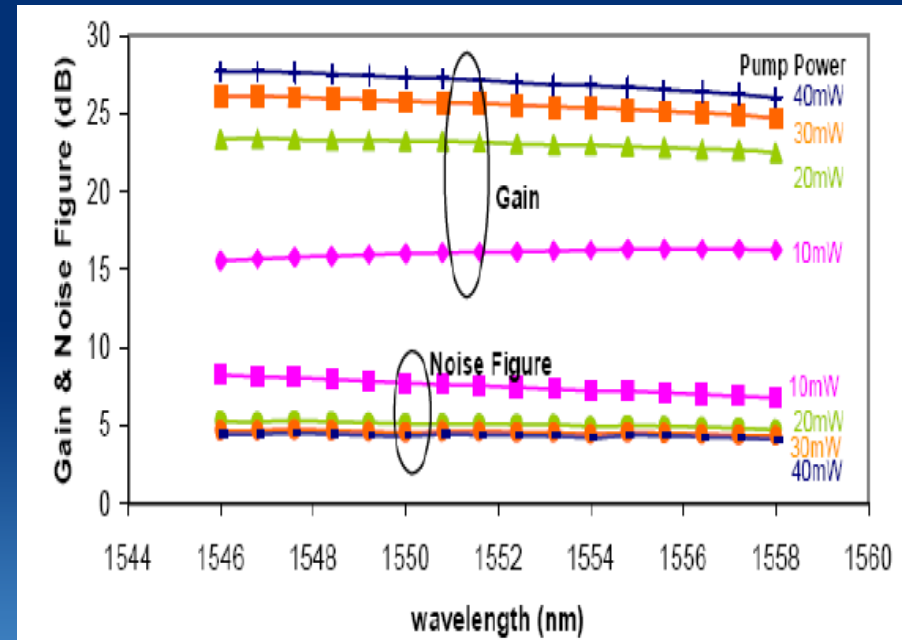
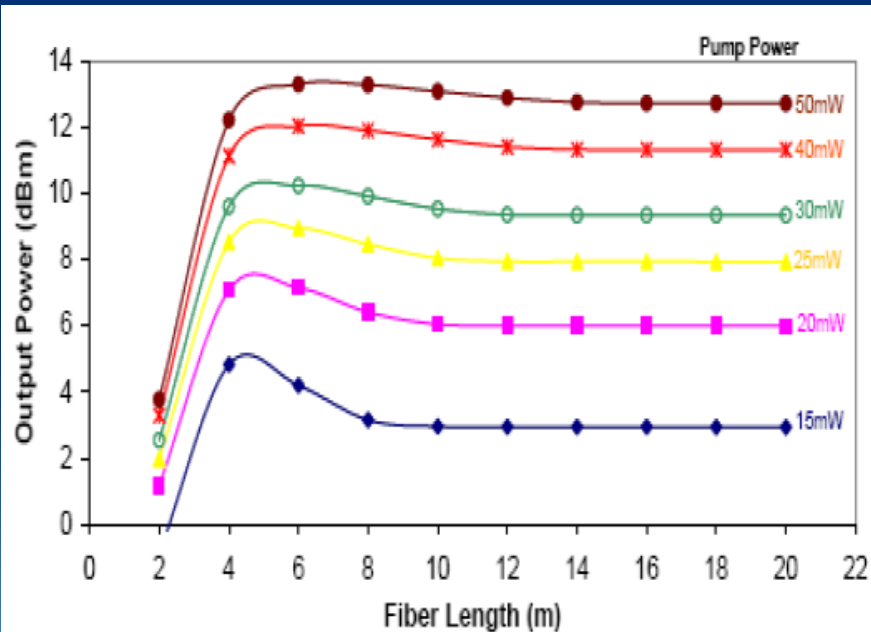
Inline Dicon gain flattening filter



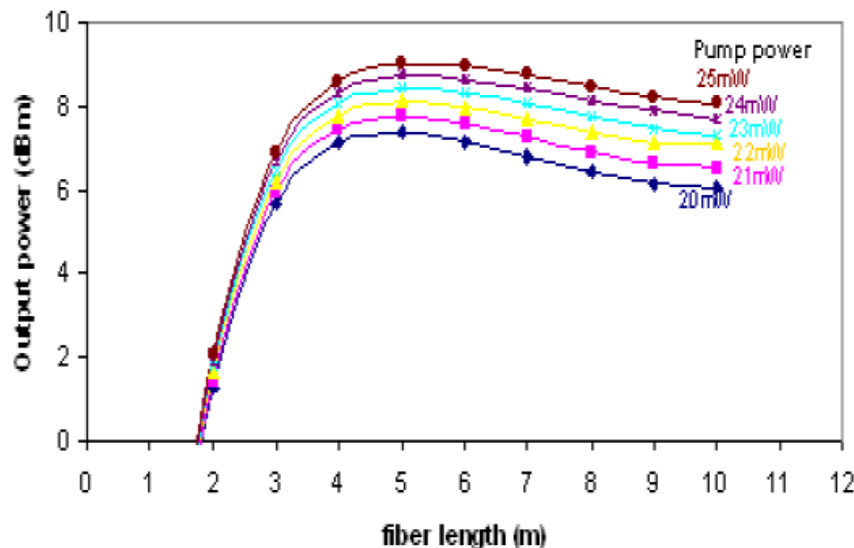
# Class Exercise and Homework



# Class Exercise and Homework



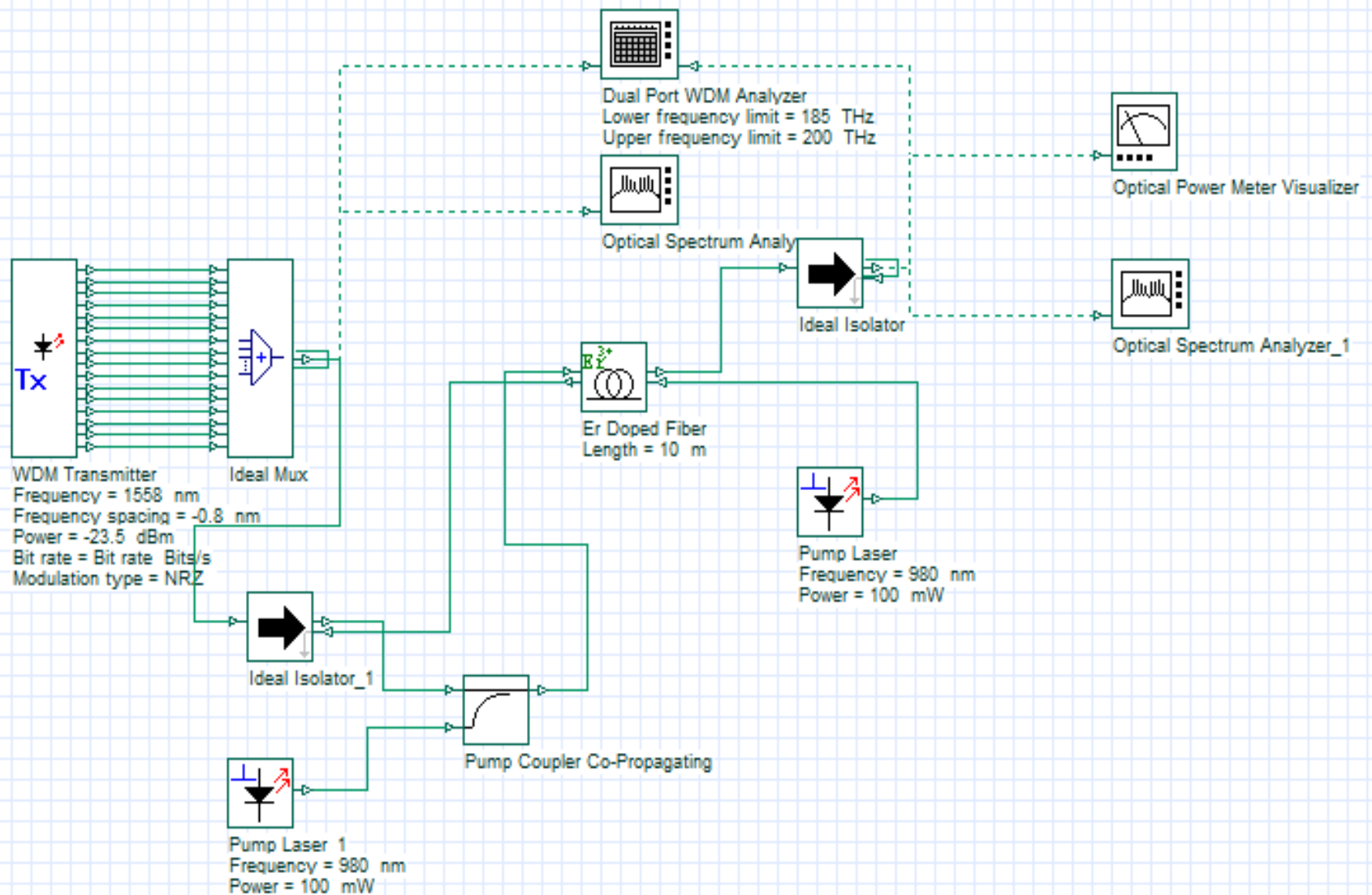
- Optimum Fiber Length~5m



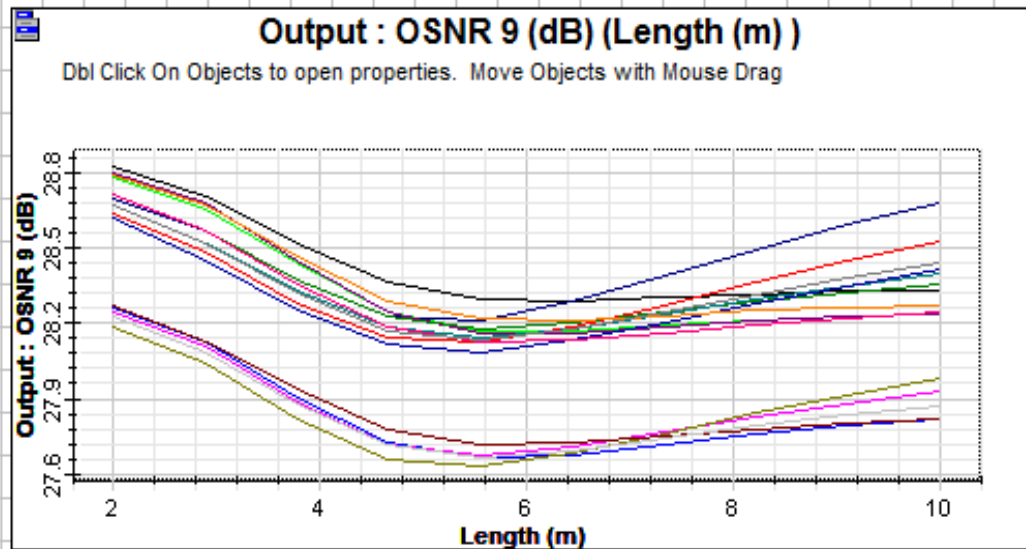
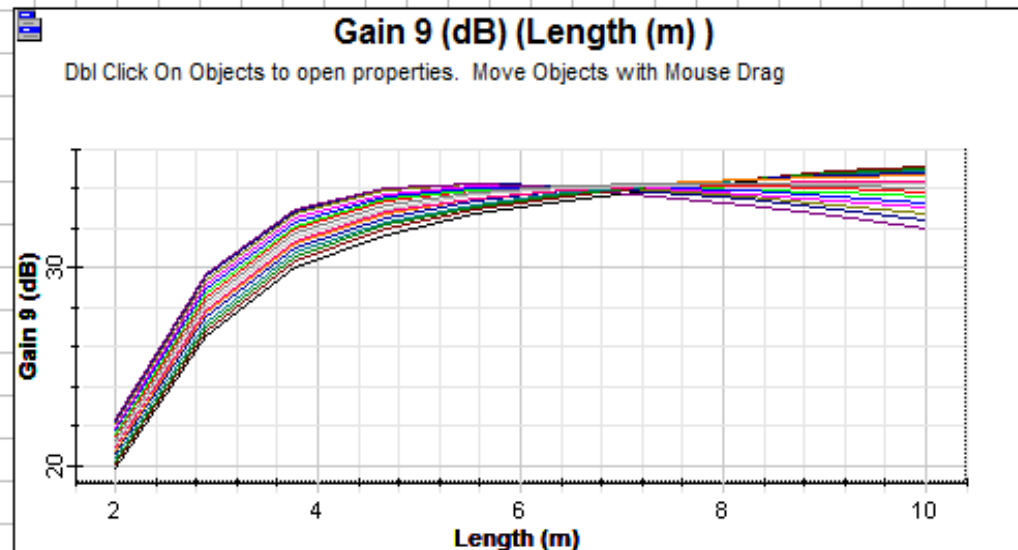
# Class Exercise

- Task
  - Change to Two-pumping scheme
  - Observe
    - Gain v.s. length
    - Noise v.s. length
    - Gain v.s. wavelength
  - Find optimum length
  - Gain and NF at the optimum length
  - Gain saturation

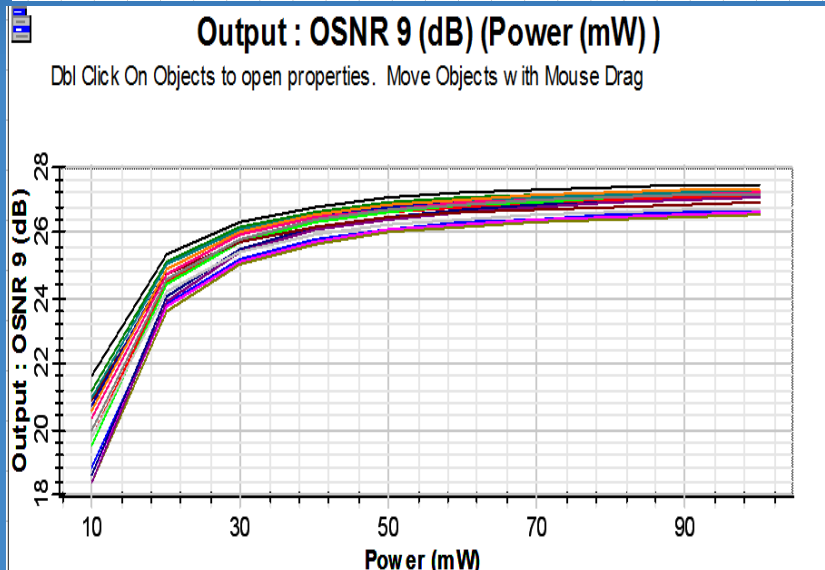
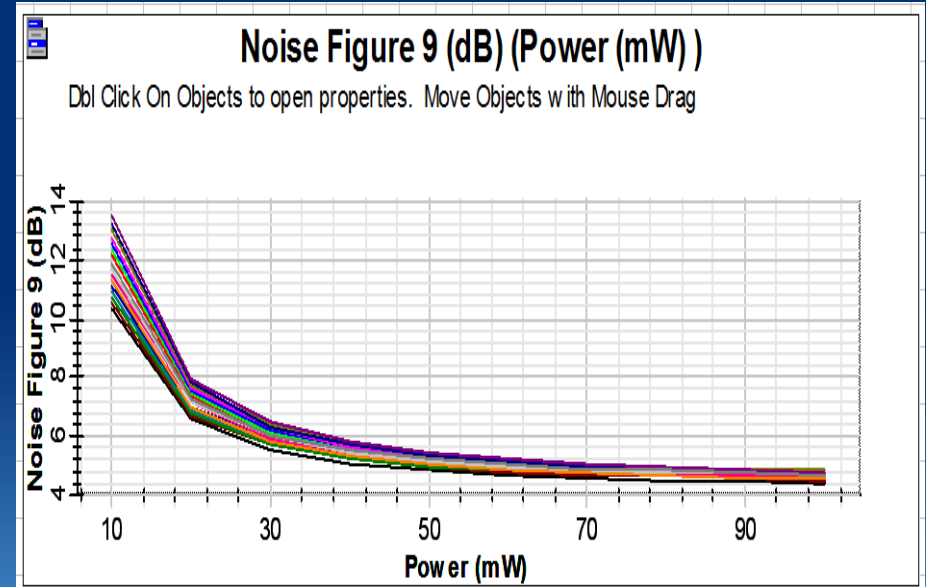
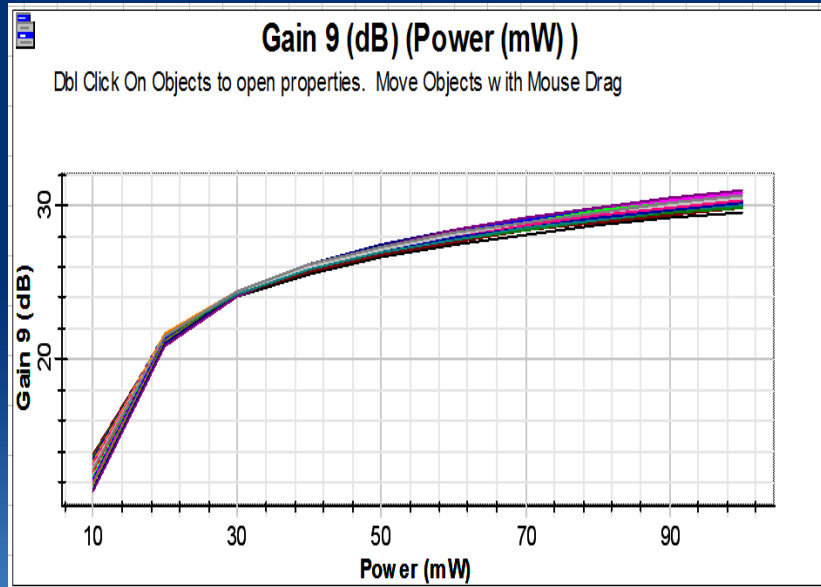
# Bi-directional pumping



# Searching for optimal fiber length



# For fixed fiber length=5m, Gain, OSNR, noise figure v.s. pumping power



**Counter-propagating Pumping only**



Observe gain saturation:  
co-pumping 100mW, input sweep from -20dBm to 10dBm

